

ASYMPTOTIC DISTRIBUTIONS FOR THE STEIN ESTIMATOR IN MULTIPLE REGRESSION

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SUMMARY

The asymptotic distribution, mean and variance of the Stein estimator is derived using a perturbation method of Fujikoshi (1968).

The asymptotic distribution of the quadratic loss function is also derived and the risk of the Stein and maximum likelihood estimators are compared.

1. Introduction

Since the exact distribution of the Stein estimator in multiple regression is not known and may be of interest to people studying confidence sets, it is the purpose of this paper to derive the asymptotic distribution, mean and variance of the Stein estimator as well as the marginal distributions.

$$\text{Let } X(p \times n) = \begin{bmatrix} \underline{X}^{(1)} \\ \underline{X}^{(2)} \end{bmatrix} \quad \begin{matrix} (1 \times n) \sim N(0; \Sigma \otimes I_n) \\ ((p-1) \times n) \end{matrix}$$

where

$$(1.1) \quad \Sigma = \begin{bmatrix} \sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}$$

It follows that

$$(1.2) \quad E(\underline{X}_i^{(1)} / \underline{X}_{-i}^{(2)}) = \underline{\beta}_{-i} \underline{X}_{-i}^{(2)}$$

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where $X_i^{(1)}$ and $X_i^{(2)}$ ((p-1)x1) are the i-th observations of $\underline{X}^{(1)}$ and $\underline{X}^{(2)}$ respectively.

$$(1.3) \quad \underline{\beta} = \underline{\Sigma}_{12} \underline{\Sigma}_{22}^{-1}.$$

For estimating $\underline{\beta}$ Stein (1960) proposed the following loss function.

$$(1.4) \quad L((\underline{\beta}, \underline{\Sigma}), \hat{\underline{\beta}}) = \frac{(\hat{\underline{\beta}} - \underline{\beta})' \underline{\Sigma}_{22} (\hat{\underline{\beta}} - \underline{\beta})}{\sigma_{11.2}}$$

where

$$(1.5) \quad \sigma_{11.2} = \sigma_{11} - \underline{\Sigma}_{12} \underline{\Sigma}_{22}^{-1} \underline{\Sigma}_{21}.$$

The ML estimator, which has constant risk with the loss function shown above, is

$$(1.6) \quad \hat{\underline{\beta}}_0 = \underline{A}_{12} \underline{A}_{22}^{-1}$$

where $\underline{A} = \underline{X}\underline{X}' \sim W(\underline{\Sigma}, n)$. Stein (1960) showed that (1.6) is inadmissible when $p \geq 4$ and

Baranchik (1964) proposed the following estimator for $\underline{\beta}$:

$$(1.7) \quad \hat{\underline{\beta}}_c = (1 - c \frac{a_{11.2}}{\hat{\underline{\beta}}_0' \underline{A}_{22} \hat{\underline{\beta}}_0}) \hat{\underline{\beta}}_0$$

where $a_{11.2}$ is independent of $\hat{\underline{\beta}}_0' \underline{A}_{22} \hat{\underline{\beta}}_0$ and of $\hat{\underline{\beta}}_0$.

$$a_{11.2} = a_{11} - \underline{A}_{12} \underline{A}_{22}^{-1} \underline{A}_{21}$$

Baranchick showed that, for $0 < c < \frac{2(p-3)}{n-p+3}$, $\hat{\underline{\beta}}_c$ has everywhere lower risk with respect to the quadratic loss function than has $\hat{\underline{\beta}}_0$. As regards the choice of c , any value of c in the given interval would give an improvement over the respective maximum likelihood estimator. Stein (1960) showed that the one corresponding to $c = \frac{p-3}{n-p+3}$ is best when $\underline{\beta} = \underline{0}$ and Baranchick (1964) proved that the optimum value of c is at least $\frac{p-3}{n-p+3}$ in the general case. In the absence of any prior information regarding $\underline{\beta}$, $\frac{p-3}{n-p+3}$ seems to be a reasonable choice for c and will be used in this paper.

2. An asymptotic approximation for $\hat{\beta}_c$

From Groenewald (1978) the approximation for $\hat{\beta}_0$, the ML estimator, is given by

$$(2.1) \quad \underline{B}_n(\underline{V}) = \underline{\beta} + \frac{1}{\sqrt{n}} \underline{B} - \frac{1}{3n} \underline{K} + \frac{1}{36n^{3/2}} \underline{L} - \frac{1}{36n^2} \underline{W} + \frac{1}{108n^{5/2}} \underline{S} + O_p(n^{-3})$$

where

$$(2.2) \quad \underline{K} = 2\underline{BC} - \underline{DB}$$

$$(2.3) \quad \underline{L} = 13\underline{BC}^2 - 11\sigma_{11.2}^{-1} \underline{B}\Sigma_{22}^{-1} \underline{B}'\underline{B} - 11 \underline{DBC} + \underline{D}^2 \underline{B} + (8p+5)\underline{B}$$

$$(2.4) \quad \underline{W} = 6\underline{BC}^3 - 10\sigma_{11.2}^{-1} \underline{B}\Sigma_{22}^{-1} \underline{B}'\underline{BC} - 7\sigma_{11.2}^{-1} \underline{BC}\Sigma_{22}^{-1} \underline{B}'\underline{B}$$

$$(2.5) \quad \underline{S} = -3\underline{B}(\text{trD}+\text{trC})+6(2p+1)\underline{BC}-7\underline{DBC}^2 + 5\sigma_{11.2}^{-1} \underline{DB}\Sigma_{22}^{-1} \underline{B}'\underline{B}-4(p+1)\underline{DB}+\underline{D}^2 \underline{BC} - 7\underline{BC}^4-11\sigma_{11.2}^{-1} \underline{BC}^2\Sigma_{22}^{-1} \underline{B}'\underline{B}-19\sigma_{11.2}^{-1} \underline{B}\Sigma_{22}^{-1} \underline{B}'\underline{BC}^2 - 19\sigma_{11.2}^{-1} \underline{BC}\Sigma_{22}^{-1} \underline{B}'\underline{BC}+11\sigma_{11.2}^{-2} \underline{B}\Sigma_{22}^{-1} \underline{B}'\underline{B}\Sigma_{22}^{-1} \underline{B}'\underline{B} + 12\sigma_{11.2}^{-1} \underline{DBC}\Sigma_{22}^{-1} \underline{B}'\underline{B}-11\underline{DBC}^3+14\sigma_{11.2}^{-1} \underline{DB}\Sigma_{22}^{-1} \underline{B}'\underline{BC} + (p+1)(8p+5)\underline{B}-15\underline{BC}(\text{trD}+\text{trC})+3(11p+4)\underline{BC}^2 - (19p+16)\sigma_{11.2}^{-1} \underline{B}\Sigma_{22}^{-1} \underline{B}'\underline{B}+2\underline{D}^2 \underline{BC}^2+(p+1)\underline{D}^2 \underline{B} - (19p+16)\underline{DBC}+3\underline{DB}(\text{trD}+\text{trC})-2\sigma_{11.2}^{-1} \underline{DB}\Sigma_{22}^{-1} \underline{B}'\underline{DB}$$

and where

$$(2.6) \quad \underline{B}(1 \times p-1) \sim N(0; \sigma_{11.2}^{-1} \Sigma_{22}^{-1})$$

$$(2.7) \quad C(p-1 \times p-1) \sim N(0; \Sigma_{22} \otimes \Sigma_{22}^{-1} + (I \otimes I)_{(p-1)})$$

$$(2.8) \quad D(1 \times 1) \sim N(0; 2)$$

with $(I \otimes I)_{(p-1)}$ a $(p-1)$ -row permuted matrix and $\underline{B}$, $\underline{C}$ and $\underline{D}$ independent of each

other.

The approximation (2.1) is needed to find an asymptotic distribution for $\hat{\beta}_0$ and is obtained by approximating a central Wishart variate by a symmetric normal variate, a multivariate analogue of the Fisher-Cornish (1960) expansion, (Nel and Groenewald (1977), Nel (1978)), i.e. $\frac{1}{n}A$ is approximated by $\frac{1}{n}U$ where

$$(2.9) \quad \frac{1}{n}U = \Sigma + \frac{1}{\sqrt{n}}V + \frac{1}{3n}(\Gamma - (p+1)\Sigma) \\ + \frac{1}{n^{3/2}}(\Delta - 3\Sigma \text{tr} \Sigma^{-1}V - (4p+7)V)$$

and

$$(2.10) \quad V = V' \sim N(O; K_p' (2\Sigma \otimes \Sigma) K_p)$$

means that V has normally distributed elements with mean O and that the upper triangular part of V regarded as a vector has covariance matrix with elements $\text{Var}(V_{ii}) = 2\sigma_{ii}^2$, $\text{Var}(V_{ij}) = \sigma_{ii}\sigma_{jj} + \sigma_{ij}^2$ and $\text{Cov}(V_{ij}, V_{kl}) = \sigma_{ik}\sigma_{jl} + \sigma_{il}\sigma_{jk}$

$$(2.11) \quad \Gamma = V\Sigma^{-1}V$$

$$(2.12) \quad \Delta = V\Sigma^{-1}V\Sigma^{-1}V$$

The approximation to $\hat{\beta}_0$ is then given by $B_n(V) = (\frac{1}{n}U_{12})'(\frac{1}{n}U_{22})^{-1}$ where $\frac{1}{n}U_{12}$ and $\frac{1}{n}U_{22}$ are the appropriate submatrices from (2.9). Equation (2.1) follows after expansion.

The relationship between V and B , C and D in (2.6) to (2.8) is

$$(2.13) \quad B = (V_{12} - \beta V_{22})\Sigma_{22}^{-1}$$

$$(2.14) \quad C = V_{22}\Sigma_{22}^{-1}$$

$$(2.15) \quad D = (V_{11} - \beta V_{21} - V_{12}\beta' + \beta V_{22}\beta')\sigma_{11.2}^{-1}$$

Approximating $\frac{1}{n}A_{n-21}$ by the corresponding submatrix in (2.9), an approximation for $\frac{1}{n}\hat{\beta}_0 A_{22}\hat{\beta}_0'$ is found and the approximation for $\frac{1}{n}\hat{\beta}_0 A_{22}\hat{\beta}_0'$ is given by

$$(2.16) \quad H = B_n(V)(\frac{1}{n}B_n(V)U_{21})^{-1}$$

Let

$$(2.17) \quad c^* = ca_{11.2} = c\sigma_{11.2}^{\nu}$$

where ν is a χ^2 variable with $n-p+1$ degrees of freedom and

$$(2.18) \quad c = \frac{p-3}{n-p+3}.$$

Then $\hat{\beta}_c$ in (1.7) can be approximated by

$$(2.19) \quad \underline{A}_n(V) = \underline{B}_n(V) - \frac{1}{n} c^* \underline{H}$$

and

$$(2.20) \quad \begin{aligned} \underline{Z}_c &= \sqrt{n} (\underline{A}_n(V) - \underline{\beta}) \\ &= \underline{B} - \frac{1}{3\sqrt{n}} (\underline{K} + \frac{3c^*}{b} \underline{\beta}) + \frac{1}{36n} (\underline{L} - \frac{36c^*}{b} \underline{\tilde{L}}) \\ &\quad - \frac{1}{36n^{3/2}} (\underline{W} + \frac{12c^*}{b} \underline{\tilde{W}}) + \frac{1}{108n^2} (\underline{S} - \frac{3c^*}{b} \underline{\tilde{S}}) \\ &\quad + O_p(n^{-5/2}) \end{aligned}$$

where

$$(2.21) \quad \underline{b} = \underline{\beta} \underline{\Sigma}_{21}$$

$$(2.22) \quad \underline{\tilde{L}} = \underline{B} - \frac{1}{b} \underline{K}^* \underline{\beta}$$

$$(2.23) \quad \underline{\tilde{W}} = \frac{3}{b^2} \underline{K}^{*2} \underline{\beta} - \frac{1}{b} \underline{L}^* \underline{\beta} - \frac{3}{b} \underline{K}^* \underline{B} - \underline{K}$$

$$(2.24) \quad \begin{aligned} \underline{\tilde{S}} &= \underline{L} + \frac{12}{b} \underline{K}^* \underline{K} + \frac{36}{b^2} \underline{K}^{*2} \underline{B} - \frac{12}{b} \underline{L}^* \underline{B} - \frac{1}{b} \underline{W}^* \underline{\beta} \\ &\quad + \frac{24}{b^2} \underline{K}^* \underline{L}^* \underline{\beta} - \frac{36}{b^2} \underline{K}^{*3} \underline{\beta} \end{aligned}$$

with

$$(2.25) \quad \underline{K}^* = \underline{\beta} \underline{V}_{21} + \underline{B} \underline{\Sigma}_{21}$$

$$(2.26) \quad \underline{L}^* = \underline{\beta} \underline{\Gamma}_{21} - (p+1) \underline{\beta} \underline{\Sigma}_{21} + 3 \underline{B} \underline{V}_{21} - \underline{K} \underline{\Sigma}_{21}$$

$$(2.27) \quad W^* = \frac{\beta \Delta}{-21} - 3 \frac{\beta \Sigma}{-21} (\text{tr} \Sigma^{-1} V) - (4p+7) \frac{\beta V}{-21} \\ + 12 \frac{B \Gamma}{-21} - 12(p+1) \frac{B \Sigma}{-21} - 12 \frac{K V}{-21} + \frac{L \Sigma}{-21}$$

3. The distribution of Z_{-c}

Theorem 3.1

Let $\hat{\beta}_{-c}(1xp)$ be given by (1.7). Then the asymptotic characteristic function of Z_{-c} , an approximation to $Z_{-c}^* = \sqrt{n}(\hat{\beta}_{-c} - \beta)$, is

$$(3.1) \quad \phi_{Z_{-c}}(\underline{T}) = \exp(-\frac{1}{2} \Lambda) \left\{ 1 + \frac{i}{\sqrt{n}} h_1(\underline{T}) + \frac{1}{n} h_2(\underline{T}) + \rho_n(\underline{T}) \right\}$$

where $n^{3/2} |\rho_n(\underline{T})|$ is bounded for all $\underline{T}$,

$$(3.2) \quad \Lambda = \sigma_{11.2} \underline{T} \Sigma_{22}^{-1} \underline{T}'$$

and

$$(3.3) \quad h_1(\underline{T}) = - \frac{p-3}{b} \sigma_{11.2} \beta \underline{T}'$$

$$(3.4) \quad h_2(\underline{T}) = \frac{1}{36} \left\{ \left(\frac{36(p-3)}{b} \sigma_{11.2} - 26p-4 \right) \Lambda + 29 \Lambda^2 \right. \\ \left. - \frac{36(p-3)(p-1)}{b^2} \sigma_{11.2}^2 (\beta \underline{T}')^2 \right\}$$

where $\beta \neq 0$ and b given in (2.21).

The density function of Z_{-c} is

$$(3.5) \quad f(Z_{-c}) = (2\pi)^{-\frac{p-1}{2}} \sigma_{11.2}^{-\frac{p-1}{2}} |\Sigma_{22}|^{1/2} \exp(-\frac{1}{2} \Lambda) \left\{ 1 - \frac{1}{\sqrt{n}} g_1(Z_{-c}) \right. \\ \left. + \frac{1}{n} g_2(Z_{-c}) + O_p(n^{-3/2}) \right\}$$

where

$$(3.6) \quad \Delta = \sigma_{11.2}^{-1} \underline{Z}_{-c} \Sigma_{22}^{-1} \underline{Z}'_{-c}$$

and

$$(3.7) \quad g_1(\underline{Z}_{-c}) = \frac{p-3}{b} \underline{Z}_{-c} \Sigma_{21}$$

$$(3.8) \quad g_2(\underline{Z}_{-c}) = \frac{1}{36} \{ (p-1)(3p+25) - \left(\frac{36(p-3)}{b} \sigma_{11.2} \right. \\ \left. + 2(16p+27) \Delta + 29\Delta^2 \right. \\ \left. + \frac{36}{b^2} (p-3)(p-1) (\underline{Z}_{-c} \Sigma_{21})^2 \}$$

Proof:

$$(3.9) \quad \phi_{\underline{Z}_{-c}}(\underline{T}) = E(\text{etr } i \underline{Z}_{-c} \underline{T}') \\ = E \{ (\text{etr } i \underline{B} \underline{T}') f(\underline{B}) \}$$

where $\underline{B}$ is given in (2.6) and $f(\underline{B})$ is the expression when the other terms in (2.20) are expanded. This reduces to

$$(3.10) \quad \phi_{\underline{Z}_{-c}}(\underline{T}) = \exp(-\frac{1}{2} \sigma_{11.2} \underline{T} \Sigma_{22}^{-1} \underline{T}') E \{ f(\underline{B}^*) \}$$

where

$$\underline{B}^* \sim N(i\sigma_{11.2} \underline{T} \Sigma_{22}^{-1}; \sigma_{11.2} \Sigma_{22}^{-1})$$

After taking expectations, (Groenewald (1978)), (3.1) follows. The density function follows after applying the inverse theorem.

Corollary 3.1

The marginal density function of Z_j , the j -th element of $\underline{Z}_{-c}$, is

$$(3.11) \quad f_{Z_j}(Z) = \frac{1}{\sqrt{2\pi} \sqrt{\sigma_{11.2} r_{jj}}} \exp\left(-\frac{1}{2} \frac{Z^2}{\sigma_{11.2} r_{jj}}\right) \left\{ 1 + \frac{1}{\sqrt{n}} g_1(Z) \right. \\ \left. + \frac{1}{n} g_2(Z) + O_p(n^{-3/2}) \right\}$$

where

$$(3.12) \quad g_1(Z) = -\frac{p-3}{b} \frac{\beta_j}{r_{jj}} Z$$

$$(3.13) \quad g_2(Z) = \frac{1}{36} \{83-26p+\sigma_{11.2}^k + (\frac{26p-170}{\sigma_{11.2}} - k) \frac{Z^2}{r_{jj}} + \frac{29Z^4}{\sigma_{11.2}^2 r_{jj}^2}\}$$

with

$$(3.14) \quad k = \frac{36(p-3)}{b} (1 - \frac{(p-1)}{b} \frac{\beta_j^2}{r_{jj}})$$

where β_j is the j -th element of $\underline{\beta}(1 \times p-1)$ and r_{jj} is the j,j -th element of Σ_{22}^{-1} .

It follows that

$$(3.15) \quad E(Z_j) = -\sqrt{\frac{1}{n}} \frac{p-3}{b} \sigma_{11.2} \beta_j + O(n^{-3/2})$$

and

$$(3.16) \quad \text{var}(Z_j) = \sigma_{11.2}^2 r_{jj} + \frac{1}{9n} \{(13p+2)\sigma_{11.2}^2 r_{jj} - \frac{9(p-3)}{b} (2r_{jj} - \frac{p+1}{b} \beta_j^2)\} + O(n^{-2})$$

4. Asymptotic approximation for the quadratic loss function

Stein (1960) proved that the loss function (1.4) is invariant under a transformation that reduces the covariance matrix to

$$(4.1) \quad \Sigma = \begin{bmatrix} 1 + \underline{\beta}\underline{\beta}' & \underline{\beta} \\ \underline{\beta}' & I_{p-1} \end{bmatrix}$$

so that $\sigma_{11.2} = 1$ and the only unknown parameter is now $\underline{\beta}$.

Consider the quadratic loss function

$$(4.2) \quad Y_c = n(\underline{A}_n(V) - \underline{\beta})(\underline{A}_n(V) - \underline{\beta})' = \underline{Z}_c \underline{Z}_c'$$

which is the approximation to

$$(4.3) \quad Y_c^* = n(\hat{\underline{\beta}}_c - \underline{\beta})(\hat{\underline{\beta}}_c - \underline{\beta})' = \underline{Z}_c^* \underline{Z}_c^{*'}.$$

From (2.20) it follows that

$$\begin{aligned}
 (4.4) \quad Y_c &= \underline{BB}' - \frac{2}{3\sqrt{n}} \underline{PB}' + \frac{1}{18n} (\underline{QB}' + 2\underline{PP}') \\
 &\quad - \frac{1}{54n^{3/2}} (3\underline{RB}' + \underline{PQ}') + \frac{1}{1296n^2} (24\underline{NB}' + 24\underline{PR}' + \underline{QQ}') \\
 &\quad + O_p(n^{-5/2})
 \end{aligned}$$

where

$$\begin{aligned}
 (4.5) \quad \underline{P} &= \underline{K} + \frac{3c^*}{b} \underline{\beta} \\
 \underline{Q} &= \underline{L} - \frac{36c^*}{b} \underline{\tilde{L}} \\
 \underline{R} &= \underline{W} + \frac{12c^*}{b} \underline{\tilde{W}} \\
 \underline{N} &= \underline{S} - \frac{3c^*}{b} \underline{\tilde{S}}
 \end{aligned}$$

This expansion of the loss function of the Stein estimator is again necessary in order to obtain an asymptotic distribution. With Σ having the structure (4.1) it follows that

$$\underline{BB}' \sim \chi_{p-1}^2$$

and, taking expected values, (Groenewald (1978)), the risk is given by

$$\begin{aligned}
 (4.6) \quad E(Y_c) &= p-1 + \frac{1}{n} \{p(p-1) - \frac{1}{b}(p-3)^2\} + \frac{1}{n^2} \left\{ \frac{1}{1296} (773p^3 \right. \\
 &\quad \left. - 704p^2 - 137p + 1508) - \frac{2}{b} p(p-3)^2 \right. \\
 &\quad \left. + \frac{1}{b^2} (p-3)(p^2 - 2p + 21) \right\} + O(n^{-3})
 \end{aligned}$$

and

$$(4.7) \quad \text{var}(Y_c) = 2(p-1) + \frac{2}{n} \{(p-1)(3p+1) - \frac{2}{b}(p-3)^2\}$$

$$\begin{aligned}
& + \frac{1}{2} \left\{ \frac{1}{324} (176p^4 + 3829p^3 + 14p^2 - 2545p - 34) \right. \\
& - \frac{1}{36b} (p-3) (319p^3 - 745p^2 - 1059p + 629) \\
& \left. + \frac{4}{b^2} (p-3) (10p^2 - 23p + 27) \right\} + O(n^{-3})
\end{aligned}$$

The quadratic loss function for the ML estimator, $\hat{\underline{\beta}}_0$, is given by

$$(4.8) \quad Y_0^* = n(\hat{\underline{\beta}}_0 - \underline{\beta})(\hat{\underline{\beta}}_0 - \underline{\beta})'$$

with the approximation

$$(4.9) \quad Y_0 = n(\underline{B}_n(V) - \underline{\beta})(\underline{B}_n(V) - \underline{\beta})'$$

where $\underline{B}_n(V)$ is given by (2.1).

Then

$$(4.10) \quad E(Y_0) = p-1 + \frac{1}{n} \{p(p-1)\} + \frac{1}{2} \left\{ \frac{1}{1296} (773p^3 - 704p^2 - 137p + 1508) \right\} + O(n^{-3})$$

and

$$\begin{aligned}
(4.11) \quad \text{var}(Y_0) &= 2(p-1) + \frac{2}{n} (p-1) (3p+1) + \frac{1}{324n^2} (176p^4 + 3829p^3 \\
&\quad + 14p^2 - 2545p - 34) + O(n^{-3})
\end{aligned}$$

From Stein (1960):

$$(4.12) \quad E(Y_0^*) = \frac{n(p-1)}{n-p}.$$

5. The distribution of Y_c

Theorem 5.1

The asymptotic characteristic function of Y_c defined in (4.4) is given by

$$(5.1) \quad \phi_{Y_c}(t) = (1-2it)^{-\frac{p-1}{2}} \{1 + \frac{1}{n} h_1(t) + \rho_n(t)\}$$

with $n^{-2} |\rho_n(t)|$ bounded for all t , independent of n , and

$$(5.2) \quad h_1(t) = \frac{1}{18} \{(p-1)(29p+11)(1-2it)^{-1}it + (p^2-1)(1-2it)^{-2}(40(it)^2-11it) + \frac{18}{b}(p-3)^2it + \frac{36}{b}(p-3)^2(1-2it)^{-1}((it)^2-it)\}$$

with $b = \underline{\beta}\beta'$ and $\underline{\beta} \neq \underline{0}$

The density function of Y_c is given by

$$(5.3) \quad f_{Y_c}(Y_c) = \frac{1}{\Gamma(\frac{p-1}{2})2^{\frac{p-1}{2}}} e^{-\frac{Y_c}{2}} Y_c^{\frac{p-1}{2}-1} \{1 + \frac{1}{n} g_1(Y_c) + O(n^{-2})\}$$

where

$$(5.4) \quad g_1(Y_c) = \frac{1}{4} \{-(p-1)^2 + \frac{2}{b}(p-3)^2 - 2(1 + \frac{(p-3)^2}{b(p-1)})Y_c + Y_c^2\}$$

Proof:

$$(5.5) \quad \begin{aligned} \phi_{Y_c}(t) &= E(\exp i Y_c t) \\ &= E\{(\exp i \underline{B}\underline{B}'t)k(\underline{B})\} \end{aligned}$$

where $k(\underline{B})$ is the expansion of the further terms in (4.4).

Since $\underline{B} \sim N(0; I_{p-1})$,

$$(5.6) \quad \begin{aligned} \phi_{Y_c}(t) &= (2\pi)^{-\frac{p-1}{2}} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \exp(-\frac{1}{2}\underline{B}\underline{B}') \exp(i\underline{B}\underline{B}'t) k(\underline{B}) d\underline{B} \\ &= (1-2it)^{-\frac{p-1}{2}} E\{k(\underline{\tilde{B}})\} \end{aligned}$$

after completing the square and where $\underline{\tilde{B}}$ in $k(\underline{\tilde{B}})$ has the distribution:

$$(5.7) \quad \tilde{\underline{B}} \sim N(0; (1-2it)^{-1} I_{p-1})$$

After taking expected values (Groenewald (1978)), the result follows. Applying the inverse Fourier transformation:

$$(5.8) \quad f_{Y_c}(Y_c) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \phi_{Y_c}(t) \exp(-i Y_c t) dt$$

the density function follows. (Groenewald and van der Merwe (1978)).

The density function of Y_0 , the approximation to the loss function for $\hat{\underline{\beta}}_0$, is given by

$$(5.9) \quad f_{Y_0}(Y_0) = \frac{1}{\Gamma(\frac{p-1}{2}) 2^{\frac{p-1}{2}}} e^{-\frac{Y_0}{2}} Y_0^{\frac{p-1}{2}-1} \{1 + \frac{1}{n} g_1^*(Y_0) + O(n^{-2})\}$$

where

$$(5.10) \quad g_1^*(Y_0) = -\frac{1}{2} \{(p-1)^2 + Y_0 - Y_0^2\}$$

Both Y_c and Y_0 are distributed as χ_{p-1}^2 in the limit as $n \rightarrow \infty$

6. A simulation study

Consider $A(6 \times 6) \sim W(\Sigma, n)$ where Σ has the structure (4.1) with $\underline{\beta} = [\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$. 10 000 Wishart matrices were simulated for $n = 20, 30$ and 40 and $p=6$. Then $\hat{\underline{\beta}}_0$ and $\hat{\underline{\beta}}_c$ were calculated and frequency tables obtained for the loss functions Y_0^* and Y_c^* given in (4.8) and (4.3). These frequency distributions fit the asymptotic density functions Y_0 and Y_c well, even for relatively small n .

The results are summarized in table 6.1. In the case of Y_0^* , the exact value of the risk is also given from (4.12). For the rest of the data, Y_0^* and Y_c^* indicate the simulation values, while Y_0 and Y_c indicate the approximations.

The risk and variance for the Stein estimator are in all cases smaller than that for the ML estimator. In (x) the relative frequencies of the number of times that the loss function for the Stein estimator is smaller than that of the ML estimator are given for the 10 000 simulations.

TABLE 6.1

		n=20	n=30	n=40
(i)	Exact E (Y_o^*)	7,143	6,250	5,882
(ii)	E (Y_o^*)	7,009	6,175	5,789
(iii)	E (Y_o)	6,775	6,122	5,819
(iv)	E (Y_c^*)	6,411	5,838	5,556
(v)	E (Y_c)	6,271	5,818	5,603
(vi)	var (Y_o^*)	29,20	19,93	15,96
(vii)	var (Y_o)	27,53	19,90	16,76
(viii)	var (Y_c^*)	23,76	17,15	14,48
(ix)	var (Y_c)	24,81	18,37	15,72
(x)	Rel. freq. of $Y_c^* < Y_o^*$	0,7376	0,6808	0,6576

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